

LOVELY PROFESSIONAL UNIVERSITY

School of Computer Science & Engineering

PROJECT REPORT

Deadlock Detection Visualizer

An Interactive Web-Based Educational Tool

Submitted By: Samarth Garg

Registration Number: 12324193

Roll Number: 55

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Department of Computer Science and Engineering

Lovely Professional University, Phagwara, Punjab

1. ABSTRACT

This project presents Deadlock Detection Visualizer, an interactive web-based educational tool developed using the Flask framework in Python. The primary objective is to help computer science students understand the concept of deadlock in operating systems through real-time animated visualizations of Resource Allocation Graphs (RAG), Wait-For Graphs (WFG), and the Banker's Algorithm. The application is fully containerized using Docker, deployed on AWS EC2 Free Tier, and follows a complete DevOps CI/CD pipeline using GitHub Actions.

2. INTRODUCTION

A deadlock is a situation in a multiprogramming environment where two or more processes are blocked indefinitely, each waiting for a resource held by another process in the group. Deadlocks are one of the most critical problems in Operating Systems, yet they remain abstract and difficult to grasp from textbook descriptions alone.

This project bridges that gap by providing an interactive visual simulator where students can construct scenarios, trigger deadlocks, and watch the detection algorithm execute step by step. The project also serves as a complete DevOps case study, demonstrating containerization, CI/CD automation, and cloud deployment on free-tier infrastructure.

The Four Coffman Conditions for Deadlock:

- Mutual Exclusion: At least one resource must be held in a non-shareable mode.
- Hold and Wait: A process holds at least one resource while waiting to acquire more.
- No Preemption: Resources cannot be forcibly taken; they must be released voluntarily.
- Circular Wait: A circular chain exists where each process waits for a resource held by the next.

3. OBJECTIVES

- Develop an interactive Flask web application to simulate deadlock scenarios.
- Implement Resource Allocation Graph (RAG) with real-time cycle detection.
- Implement Wait-For Graph (WFG) for process-level deadlock visualization.
- Demonstrate Banker's Algorithm for deadlock avoidance with step-by-step output.
- Containerize the application using Docker for portability and reproducibility.
- Establish a complete CI/CD pipeline using GitHub Actions for automated deployment.
- Deploy the application on AWS EC2 Free Tier with Nginx as a reverse proxy.
- Provide a student-friendly UI to enhance conceptual understanding of deadlocks.

4. TECHNOLOGY STACK

Backend:

- Python 3.11 with Flask framework for web server and API routes
- Deadlock detection logic implemented in pure Python

Frontend:

- D3.js v7 for interactive Resource Allocation and Wait-For Graph rendering
- HTML5, CSS3, and JavaScript for responsive student-friendly UI

DevOps Tools:

- Git and GitHub for version control and source code management
- Docker and Docker Compose for containerization and local development
- GitHub Actions for automated CI/CD pipeline
- AWS EC2 t2.micro (Free Tier) for cloud deployment
- Nginx as reverse proxy (port 80 to 5000)
- Docker Hub for storing and versioning Docker images

5. SYSTEM ARCHITECTURE

The application follows a layered architecture separating concerns across presentation, business logic, and infrastructure layers. A git push to the main branch triggers the entire automated chain from testing to live deployment.

Architecture Flow:

- Student Browser sends HTTP requests to Nginx on port 80
- Nginx reverse proxy forwards requests to Flask app on port 5000
- Flask processes the request using deadlock detection algorithms
- Response returned to browser; D3.js renders the visualization

CI/CD Pipeline Flow:

- Developer pushes code to GitHub main branch
- GitHub Actions triggers automatically
- Pytest runs all unit tests
- Docker image is built and tagged
- Image is pushed to Docker Hub registry
- GitHub Actions SSHs into AWS EC2 instance
- EC2 pulls the new image, stops old container, starts new one

6. CORE ALGORITHMS AND IMPLEMENTATION

6.1 Resource Allocation Graph (RAG)

A Resource Allocation Graph is a directed graph used to describe and detect deadlocks. It contains two types of nodes: Process nodes (circles) and Resource nodes (rectangles). Two types of edges exist: a Request edge (P to R) meaning process P is waiting for resource R, and an Assignment edge (R to P) meaning resource R is currently allocated to process P.

Deadlock Detection Rule: If the RAG contains a cycle AND every resource in the cycle has exactly one instance, a deadlock exists. The application uses DFS-based cycle detection with animated path highlighting to make this visible to students.

6.2 Wait-For Graph (WFG)

The Wait-For Graph is derived from the RAG by removing resource nodes and collapsing edges. If process P_i is waiting for a resource held by P_j , then a directed edge P_i to P_j exists in the WFG. A deadlock exists if and only if the WFG contains a cycle. The visualizer allows students to toggle between RAG and WFG views for better understanding.

6.3 Banker's Algorithm (Deadlock Avoidance)

Dijkstra's Banker's Algorithm is a deadlock avoidance algorithm. It simulates resource allocation to check if the system remains in a safe state before granting any request. A safe state guarantees all processes can eventually complete.

- Step 1: Define Available[], Max[], Allocation[], Need[] = Max - Allocation
- Step 2: Find a process P where Need[P] is less than or equal to Available
- Step 3: Simulate finish: add Allocation[P] back to Available; mark P as finished
- Step 4: Repeat until all processes finish (SAFE) or no such P exists (UNSAFE)

7. DEVOPS AND CLOUD DEPLOYMENT

7.1 Dockerization

The Flask application is containerized using a Dockerfile based on python:3.11-slim. This ensures the container is lightweight (approximately 128 MB) and reproducible across any environment. Docker Compose is used for local development, enabling the app to start with a single command: `docker-compose up --build`

7.2 GitHub Actions CI/CD Pipeline

A YAML workflow file at `.github/workflows/deploy.yml` defines the complete pipeline triggered on every push to the main branch. The pipeline consists of five stages:

- Stage 1 - Checkout: actions/checkout pulls the latest source code
- Stage 2 - Test: pytest runs all unit tests; pipeline stops if any test fails

- Stage 3 - Build: Docker builds a tagged image (username/deadlock-visualizer:latest)
- Stage 4 - Push: Image is pushed to Docker Hub registry
- Stage 5 - Deploy: SSH into EC2, pull new image, restart container

7.3 AWS EC2 Free Tier Deployment

The application is deployed on an AWS EC2 t2.micro instance running Ubuntu 22.04 LTS, available under AWS Free Tier (750 hours per month at no cost). Nginx acts as a reverse proxy forwarding public traffic on port 80 to the Flask container on port 5000. The Docker container is configured to auto-restart on reboot.

- Instance Type: t2.micro (1 vCPU, 1 GB RAM)
- AMI: Ubuntu Server 22.04 LTS (Free Tier Eligible)
- Storage: 8 GB gp2 EBS volume (Free Tier)
- Security Group: Inbound ports 22 (SSH), 80 (HTTP), 443 (HTTPS)
- Cost: Free within 12-month AWS Free Tier

8. PROJECT FILE STRUCTURE

- deadlock-visualizer/
- app.py -- Flask routes and deadlock algorithms
- requirements.txt -- Python dependencies
- Dockerfile -- Container build instructions
- docker-compose.yml -- Local development orchestration
- .dockerignore
- .github/
- workflows/
- deploy.yml -- GitHub Actions CI/CD pipeline
- static/
- css/style.css -- Application stylesheet
- js/visualization.js -- D3.js graph rendering logic
- templates/
- base.html -- Base HTML layout
- index.html -- Landing page
- simulator.html -- Main visualization page
- tests/
- test_app.py -- Pytest unit tests

9. APPLICATION FEATURES

- Interactive Graph Builder: Add or remove processes and resources via the UI. Assign or request resources with a single click. Graph updates in real time.
- Real-Time Deadlock Detection: DFS-based cycle detection runs on every graph change. Deadlocked nodes are highlighted with animation.
- Banker's Algorithm Simulator: Input Allocation and Max matrices. The app computes the safe sequence step by step and shows the execution order.
- Step-by-Step Explanation: Every algorithm step is narrated in a side panel with plain-English explanations designed for students.
- Presentation Mode: Pre-loaded demo scenarios with keyboard navigation, ideal for classroom demonstrations before teachers.
- Export Functionality: Export the current graph as PNG image.

10. RESULTS AND OUTCOMES

- Deadlock Detection Accuracy: 100 percent verified on multiple test cases
- Banker's Algorithm: Correct safe sequence generated for all valid inputs

- Docker Image Size: Approximately 128 MB using python:3.11-slim base
- CI/CD Pipeline Duration: Approximately 3 minutes from push to live deployment
- AWS Deployment: Live on EC2 t2.micro with zero downtime deployment
- Page Load Time: Under 1.2 seconds on Free Tier EC2 instance
- Cost: Entire infrastructure deployed at zero cost using free tiers

11. CONCLUSION

This project successfully demonstrates the intersection of core Operating Systems theory and modern DevOps engineering practices. The Deadlock Detection Visualizer provides students with an engaging, interactive medium to understand one of the most challenging topics in OS, while simultaneously serving as a practical implementation of a production-grade DevOps pipeline.

By combining Flask for backend logic, D3.js for rich visualizations, Docker for containerization, GitHub Actions for CI/CD automation, and AWS EC2 for cloud hosting, this project mirrors real-world software delivery workflows used in the industry. Every push to the main branch automatically tests, builds, and deploys the latest version, demonstrating true continuous delivery.

Future enhancements could include multi-instance resource support, deadlock recovery simulation, user authentication, and integration with monitoring tools like Prometheus and Grafana for observability.

12. REFERENCES

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